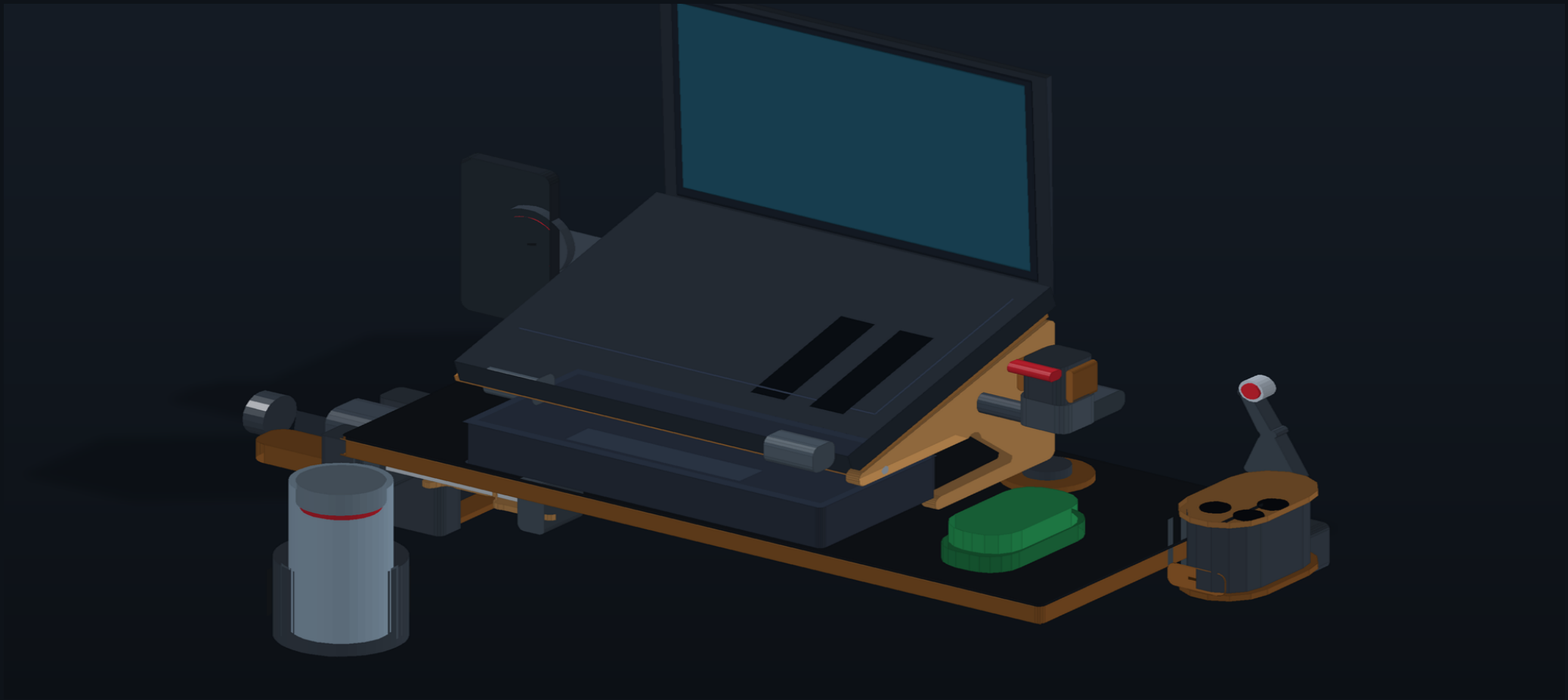
DESIGNER HYBRID V2

The 630 × 280 Desk System

Six clamp-on products, forty-three pieces, zero screws in the desk.



630 × 280 × 21
desk + mat footprint

43
pieces catalogued

6
clamp-on products

0
screws or glue in the desk

Built from Autodesk Fusion parametric models — every dimension checked by verify_fusion_scripts.py

2026

What is in this book

01 **How to read this book** 03
The three drawings on every product page, and the colour key

02 **The system at a glance** 04
One desk, six products, and where each one clamps

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What each material is doing, and the 70% filament saving

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09 **Product 4 — HexWing + Chimera Spire** 13
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10 **Product 5 — HoloBuddy** 14
The desk-top hologram, three ways to mount it

11 **Every piece, in one table** 15
All 43 parts with size, material and fabrication method

12 **Cut guides** 18
12 mm sheet, 6 mm sheet and one length of Ø8 mm rod

13 **Building it** 19
Order of operations, from the first print to the last magnet

14 **Daily use** 20
What each product does for you on a normal working day

15 **Appendix** 21
Files, names, and the checks this book depends on

Page numbers are the printed folio at the bottom right of every page.

Three drawings per product, always in the same order

A The shaded view

A product render built from the same geometry as the Fusion file. This is what the assembled product looks like on the desk. Brown is wood, grey-black is printed, bright metal is a Ø8 mm rod.

B Top / Front / Side

True orthographic projections drawn from the same geometry. Every dimension is in millimetres and can be measured with a ruler on a print-out.

C The piece list

A table naming each part exactly as it is named in Fusion, with its size, its material and how you make it. The same names appear in the exported STL / STEP files.

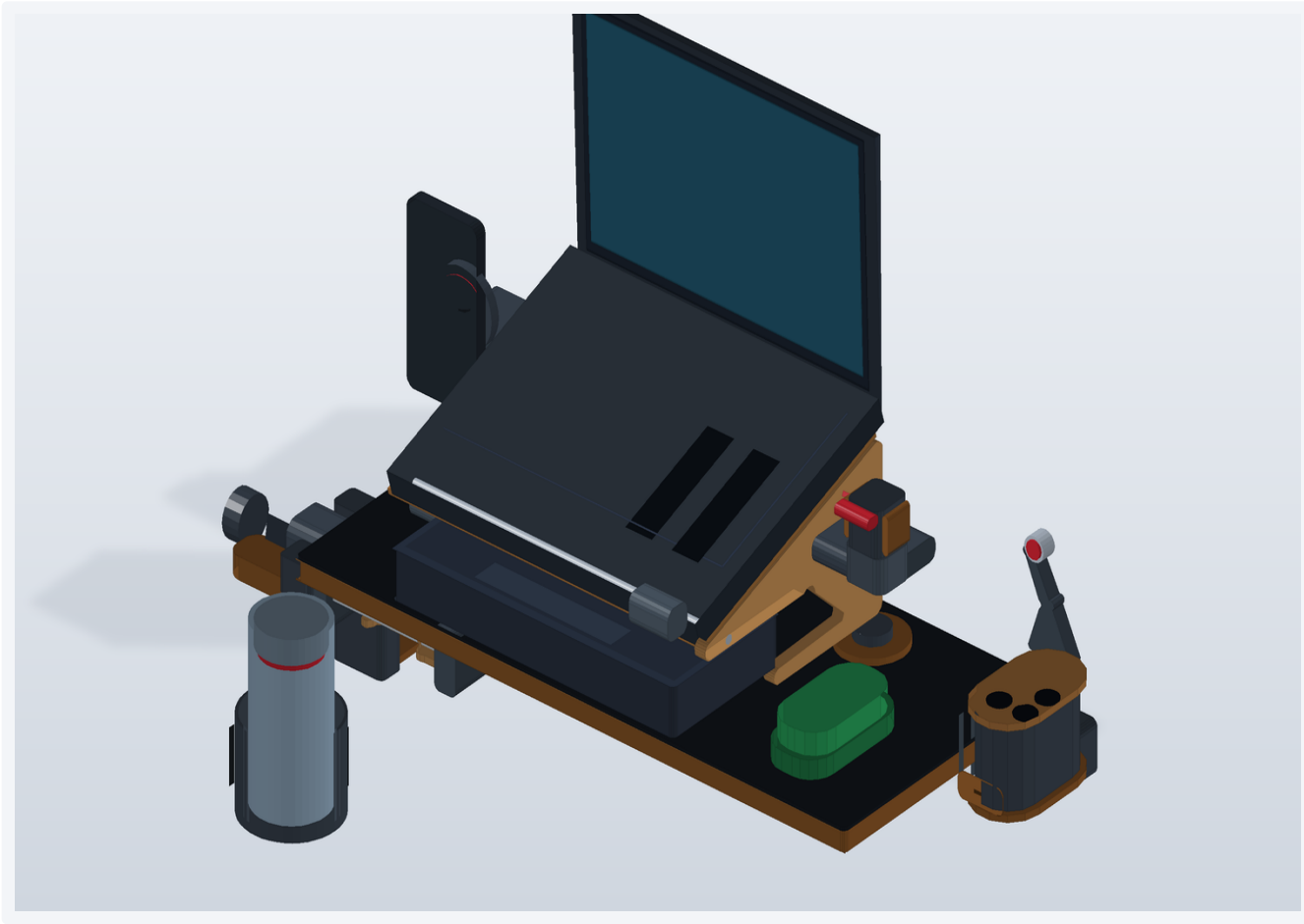
MATERIAL COLOUR KEY

- **Warm walnut / birch** — 12 mm and 6 mm sheet — arches, decks, trays, coasters
- **Brushed aluminium / steel** — Ø8 mm tie-rods, 608 bearing seats
- **Stealth matte black** — PETG precision nodes — clamps, collars, hubs
- **Cyber-Red accent** — Dovetail collars, axle collars, visors, spires
- **K9 optical glass** — The 25.4 mm beam-splitter cube in HoloBuddy

READING THE DIMENSIONS

Every length is in millimetres. A size such as 184 × 131.4 × 12 means length × height × thickness for a flat part, or X × Y × Z for an assembly. Angles are in degrees. The desk coordinate system runs X = 0...630 mm left to right, Y = 0 at the mat surface (positive up), and Z = 0 at the back wall with Z = 280 mm at the front edge you sit at. A round part is given as Ødiameter × thickness.

Where each product clamps, and the space it must leave free



Assembled, seen from the user's front-right. Products 1-5 are in place; the keyboard sits in the garage.

P0 **Fit Check Gauge** 09
Print this first

P1 **AeroBridge G16 + MagPivot S24+** 10
Laptop stand, keyboard garage, phone arm

P2 **StealthVault 280 + SideDock ULT** 11
Hide the brick, hang the headphones

P3 **HydroSling** 12
Bottle outside the desk, notebook underneath

P4 **HexWing + Chimera Spire** 13
Pens, drawer and a mouse bungee that clears the sweep

P5 **HoloBuddy** 14
A hologram in a 25.4 mm cube

CLEAR ZONES — NEVER BLOCK THESE

Knee void X 195...610 under the desk · mouse sweep X 470...628, Z 90...278 · keyboard garage 332 mm wide by 62 mm high.

TABLE	MAT	EDGE STACK	BACK WALL	USER
630 × 280 × 18 mm dark wood	630 × 280 × 3 mm	21 mm	Z = 0, flush, no rear clamp	right-handed, seated at X = 315

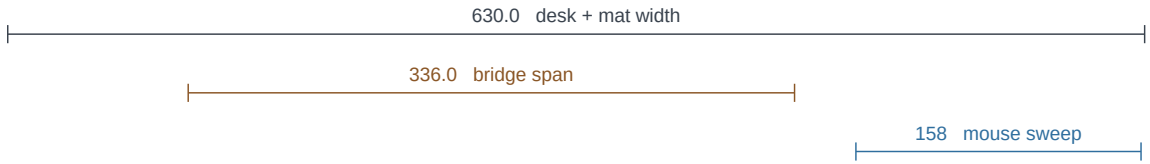
All dimensions millimetres. Part names match the exported STL / STEP files exactly.

Top plan — where everything sits

Top plan, lid and keyboard removed for clarity. Z = 0 is the back wall.



Assembled from above. Wood is brown, printed nodes grey-black, the mat near-black.



READ THIS VIEW LIKE A DRAWING

- X runs 0...630 mm left to right; Z runs 0...280 mm from the back wall to the front edge you sit at.
- Products 2 and 3 work entirely outboard: negative X or Z beyond 280 mm. They never take desk space.
- The right-hand area X 470...628, Z 90...278 is deliberately empty — that is your mouse sweep.
- The keyboard garage is the 332 mm gap between the two AeroBridge arches, X 106.5...429.5.

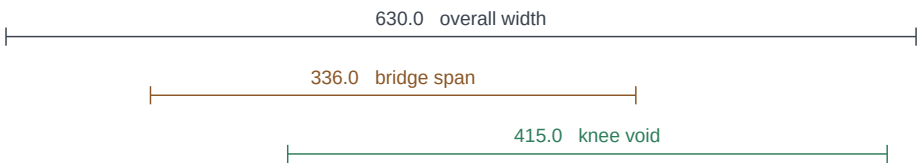
Top plan drawn from the same geometry as the Fusion model.

Front elevation — clearances you can measure

Front elevation. The 21 mm edge stack is the wood slab plus the mat.



The keyboard sits in the garage and the lid stands clear of the wall.



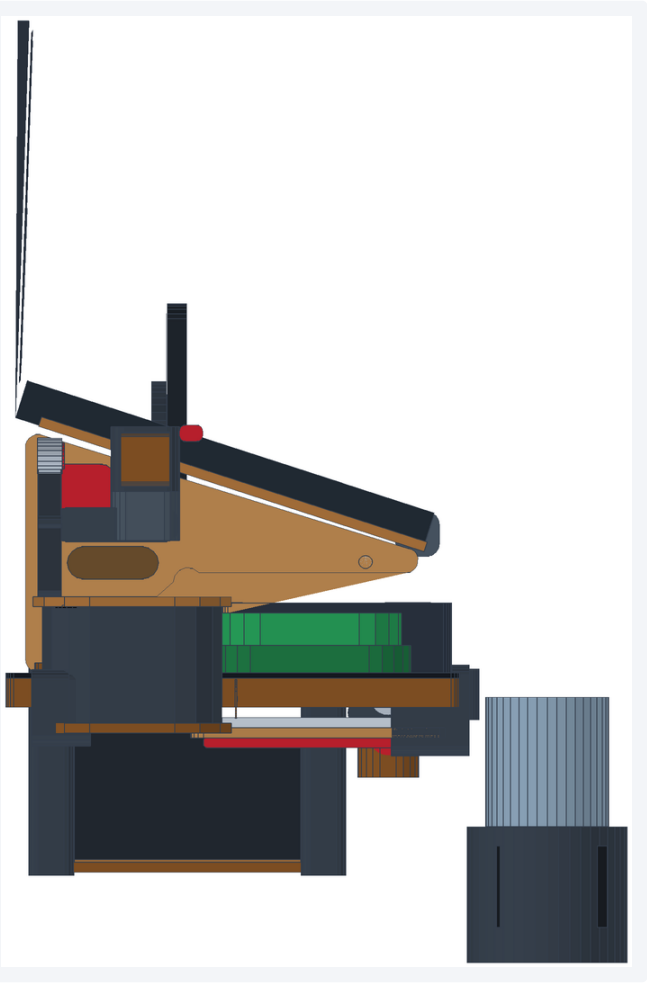
READ THIS VIEW LIKE A DRAWING

- The 62 mm gap under the bridge is the keyboard garage: the AULA F75 is 43.1 mm tall, leaving 18.9 mm of finger room.
- The 415 mm void between the left vault and the right caddy is where your knees go. Nothing hangs into it.
- Both Ø8 mm rods pass through the arches at 89 mm and 71.5 mm, and they are what keeps the bridge square.
- Every clamp descends below the slab inside its own footprint, which is what lets it grip with no screws.

Front elevation drawn from the same geometry as the Fusion model.

Right side elevation — the depth stack

Right side elevation. The desk is 280 mm deep; the bottle hangs 103.5 mm beyond it.



Depth stack from the right. The lid stands 2° off vertical, so it reads as a thin edge at the back wall.



READ THIS VIEW LIKE A DRAWING

- The laptop deck sits at 18° and the 245 mm screen stands almost vertical, so a laptop at eye level still clears the wall.
- The whole build stays inside Z = 0...280 except the bottle cage, which hangs 103.5 mm past the front edge.
- HoloBuddy floats at Y = 65...156 mm, well above the 43 mm mouse, so the sweep stays clear.
- Nothing protrudes behind Z = 0 — the desk is flush to the wall with no rear clamp at all.

THE DEPTH BUDGET

TOTAL RISE ABOVE THE MAT
406 mm top of the laptop screen

TALLEST FIXED PART
383.5 mm the top of the bottle, in the chalice

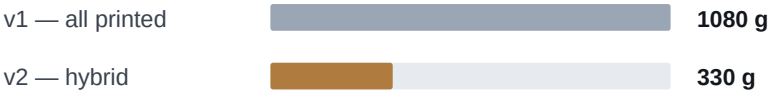
BELOW THE DESK
176 mm the bottom of the vault and the headphone saddle

DESK DEPTH USED
280 mm the full mat, including the front lip

Right side elevation drawn from the same geometry as the Fusion model.

Cheaper, stiffer, faster to make — and it looks like furniture, not a print farm

THE 70 % SAVING



Filament only. Print time falls from about 34 h to about 10.5 h.

PRINT TIME

≈ 34 h → ≈ 10.5 h

Every node fits a 200 mm bed, so one machine finishes the set overnight.

STIFFNESS

> 15 kg, < 0.2 mm sag

A 6 mm walnut deck between two Ø8 mm rods beats a 6 mm printed slab.

COST

≈ $35 in total

Walnut offcuts plus one rod cost less than the filament they replace.

LOOK

furniture, not prototype

Oiled walnut beside matte-black nodes and brushed metal reads as designed.

FINISH SEQUENCE

Sand to 320 grit → damp cloth → one coat Danish oil (or mineral oil + beeswax) → buff after 24 h.

WHAT EACH MATERIAL IS ACTUALLY DOING

Wood — 12 mm & 6 mm sheet

Carries the load and sets the look. Arches, the laptop deck, the vault shield, trays and coasters are all flat parts: a saw, a drill and sandpaper are enough. Wood is the cheapest way to make something big, warm and stiff.

Metal — Ø8 mm rod

Ties the two wood arches into one rigid bridge and gives the 608 bearings a true axle. Two 360 mm rods replace what would otherwise be a printed beam far too long for most beds, and far too flexible.

Print — 26 compact nodes

Does the jobs only printing can do: dovetail sockets held to ±0.15 mm, magnet pockets, bearing seats, cable channels and the optical chamber. Nothing is bigger than 186.5 mm, so any 200 mm printer makes the whole set.

Glass — one cube

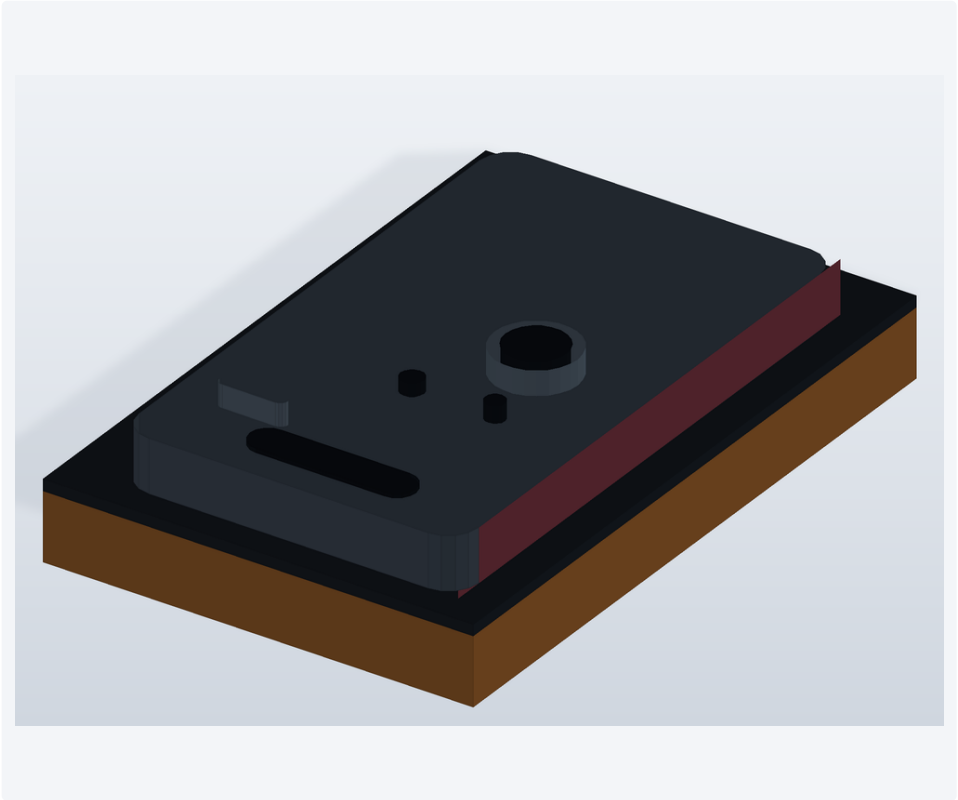
The only optical part. A 25.4 mm 50:50 K9 cube splits the OLED image so it appears to float. Bought, not made.

All dimensions millimetres. Part names match the exported STL / STEP files exactly.

PRINT THIS FIRST

P0 · Fit Check Gauge

A single 90 × 150 × 14 mm coupon that tests your printer and six pieces of hardware before you commit to the full set. If the coupon fits, every other print in this book will fit.



The coupon. Six hardware tests on one 14 mm plate, printed before anything else.

1 PIECE IN THIS PRODUCT

P0 Fit coupon	90 x 150 x 14

WHAT IT DOES FOR YOU

- Check that your printer holds ±0.15 mm on a dovetail.
- Press a 608 bearing in and feel whether the seat is right.
- Drop a 6 × 3 mm magnet into its pocket and confirm it sits flush.
- Confirm the M3 and M4 brass inserts melt in square.
- Check the DT-30 tongue slides — the same joint every product uses.
- Keep it in the spares box: it is the reference for every reprint.

PUTTING IT TOGETHER

1. Print at 0.2 mm layers with the material you intend to use, not a test spool.
2. Fit each hardware item as soon as the coupon cools — the pockets are a warm-room press fit.
3. If anything is too tight, adjust only that pocket in Fusion and re-print the coupon.

MATERIALS IN THIS PRODUCT

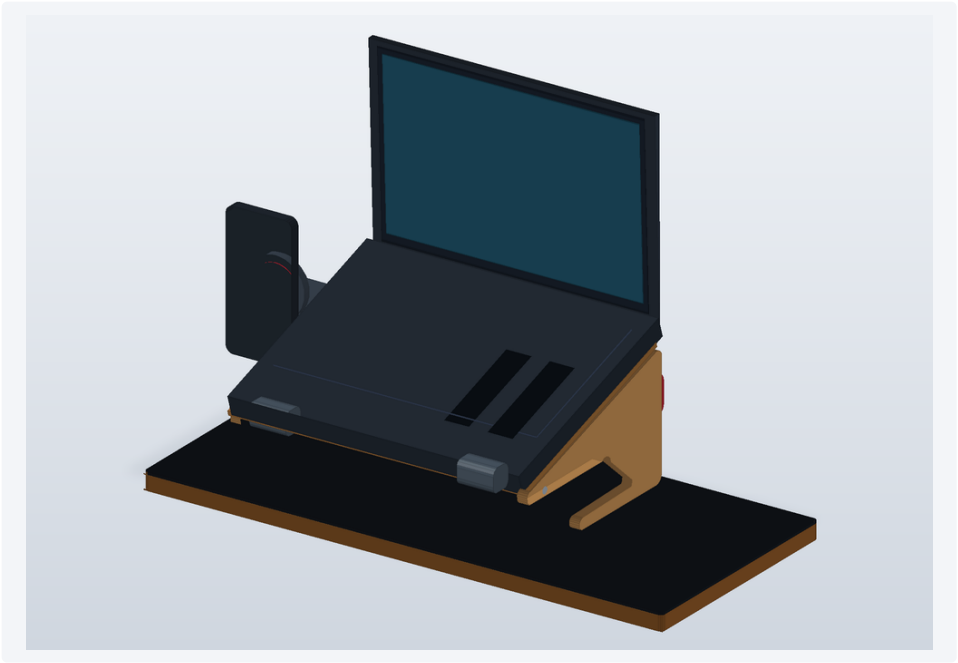
- PETG

All dimensions millimetres. Part names match the exported STL / STEP files exactly.

LAPTOP STAND, KEYBOARD GARAGE, PHONE ARM

P1 · AeroBridge G16 + MagPivot S24+

Two 12 mm walnut arches joined by two Ø8 mm aluminium rods carry a 6 mm walnut slat deck. It lifts the ROG Strix G16 to eye level at 18° and leaves a 62 mm-high garage underneath for the keyboard. A magnetic arm on the left arch holds the S24+ where you can see it.



AeroBridge with the G16 and the S24+ in place, at the 34° / 24° review angle.

11 PIECES IN THIS PRODUCT

W1A Left arch	184 x 131.4 x 12	P1D Stopper R	44 x 27.5 x 27.2
W1B Right arch	184 x 131.4 x 12	P1E DT-30 collar L	11 x 44 x 44
W1C Slat deck	360 x 182 x 6	P1E DT-30 collar R	11 x 44 x 44
M1A Rear tie-rod	Ø8.0 x 360	P1F MagPivot arm	68.8 x 38 x 60
M1B Front tie-rod	Ø8.0 x 360	P1G MagSafe disc	Ø70 x 9
P1D Stopper L	44 x 27.5 x 27.2		

WHAT IT DOES FOR YOU

- Slide the keyboard into the garage at the end of the day and the desk is instantly clear.
- The 18° deck angle and the open slats let the G16 breathe — intakes are never against a surface.
- The phone arm tilts from –8° to +62°, so it works as a video-call screen or a bedside clock.
- Two silicone-capped stoppers stop the laptop sliding without marking it.
- The right-hand DT-30 collar is where HoloBuddy (Product 5) docks.

PUTTING IT TOGETHER

1. Lay the two arches face to face, then slide both rods through. The bridge is self-squaring.
2. Drop the slat deck into the two notches — no fasteners, it locates on the arch profile.
3. Clip the stoppers onto the front deck edge and slide the phone arm onto the left rod.

MATERIALS IN THIS PRODUCT

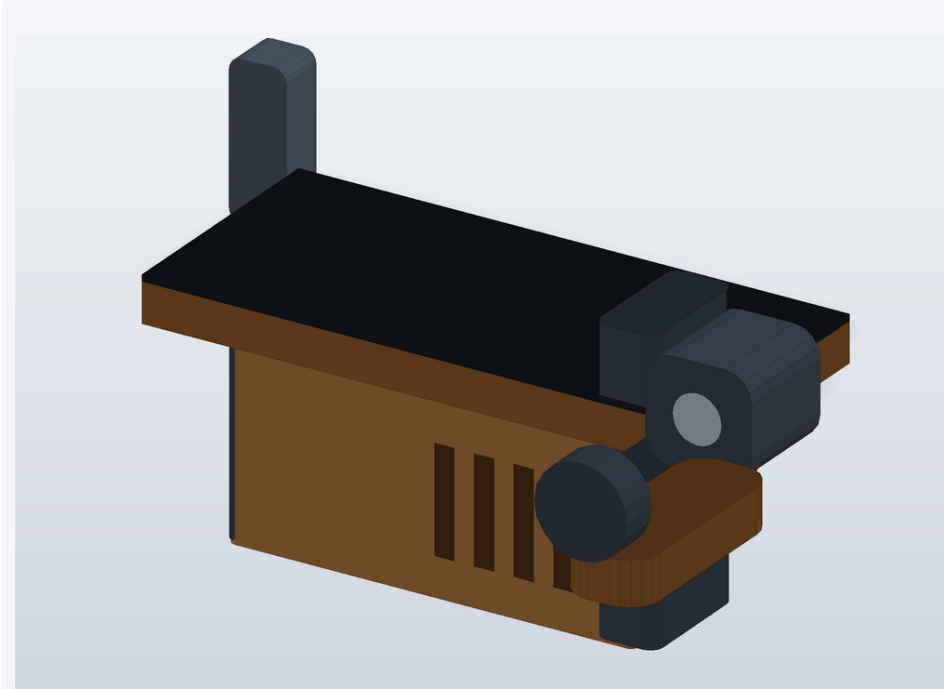
- 12 mm walnut / birch ply
- 6 mm walnut
- 6061 aluminium / 304 stainless
- TPU 95A
- PETG

All dimensions millimetres. Part names match the exported STL / STEP files exactly.

HIDE THE BRICK, HANG THE HEADPHONES

P2 · StealthVault 280 + SideDock ULT

A walnut-shielded box that clamps under the left edge and swallows the 280 W ROG power brick and its cable nest. Below it, a 608 bearing swings the Sony ULT WEAR out of the way in one 90° movement.



Vault and SideDock from the outboard side: vent slots on the shield, the 608 dock with the headphones on it, and the two clamp arches.

8 PIECES IN THIS PRODUCT

W2A	Vault floor	184 x 52 x 6	P2B	Front clamp arch	58 x 133.5 x 28
W2B	Vault shield	184 x 84 x 6	P2C	Cable spine	22 x 70 x 26
W2C	ULT saddle	99 x 38 x 20	P2D	608 bearing hub	50 x 47.5 x 48
P2A	Rear clamp arch	58 x 133.5 x 28	P2E	Axle collar	24 x 13.4 x 24

WHAT IT DOES FOR YOU

- The brick stops living on the floor, so it stops collecting dust and tripping feet.
- Nine vent slots and an open bottom keep the brick cool; it is never fully enclosed.
- Swing the headphones out to put them on, push them back and they vanish under the desk.
- Cables route through the printed spine and out of sight — one cable to the wall, the rest hidden.
- The clamp jaws carry 2 mm EVA pads, so the walnut desk is never marked.

PUTTING IT TOGETHER

1. Clamp the two printed arches over the left edge, 184 mm apart, and tighten the thumbscrews by hand.
2. Slide the walnut floor into the arches, then hook the shield on from outside.
3. Seat the 608 bearing into the hub, push the axle collar through the saddle and nip the M4 grub screw.

MATERIALS IN THIS PRODUCT

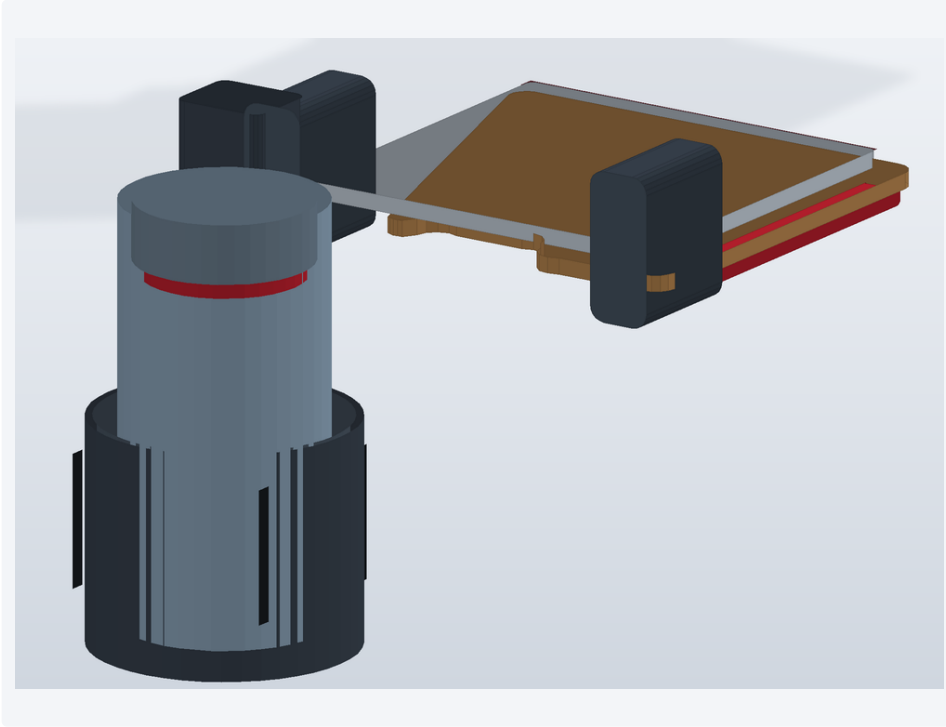
- 6 mm birch ply
- 6 mm walnut
- 20 mm walnut block
- PETG

All dimensions millimetres. Part names match the exported STL / STEP files exactly.

BOTTLE OUTSIDE THE DESK, NOTEBOOK UNDERNEATH

P3 · HydroSling

A sculpted PETG chalice hangs a 65–84 mm bottle outboard of the front-left edge, entirely off the work surface. Below the desk, a 6 mm walnut tray slides 110 mm on two DT-30 rails and holds an A5 notebook and pen.



HydroSling at 34° / 17°. The chalice hangs the bottle clear of the desk; the notebook tray sits under it.

7 PIECES IN THIS PRODUCT

W3A Bottle coaster	Ø84 x 6	P3B Right runner	24 x 58 x 48
W3B Notebook tray	172 x 162 x 6	P3C Slide cleat L	14 x 6 x 146
P3A Bottle chalice	99 x 186.5 x 121.5	P3C Slide cleat R	14 x 6 x 146
P3B Left runner	24 x 58 x 48		

WHAT IT DOES FOR YOU

- The bottle is off the desk, so a spill cannot reach the laptop — the main reason this product exists.
- The walnut coaster inside the chalice catches condensation.
- Pull the notebook tray out while you write, push it back and it disappears under the desk.
- Two tulip windows in the chalice let you see how much water is left.
- The whole thing lifts off in one piece when you need the corner of the desk.

PUTTING IT TOGETHER

1. Clamp the two runners over the front-left edge, 148 mm apart.
2. Slide the chalice body down onto the runners until it seats.
3. Drop the walnut coaster into the base, hang the tray on its cleats and check the 110 mm travel.

MATERIALS IN THIS PRODUCT

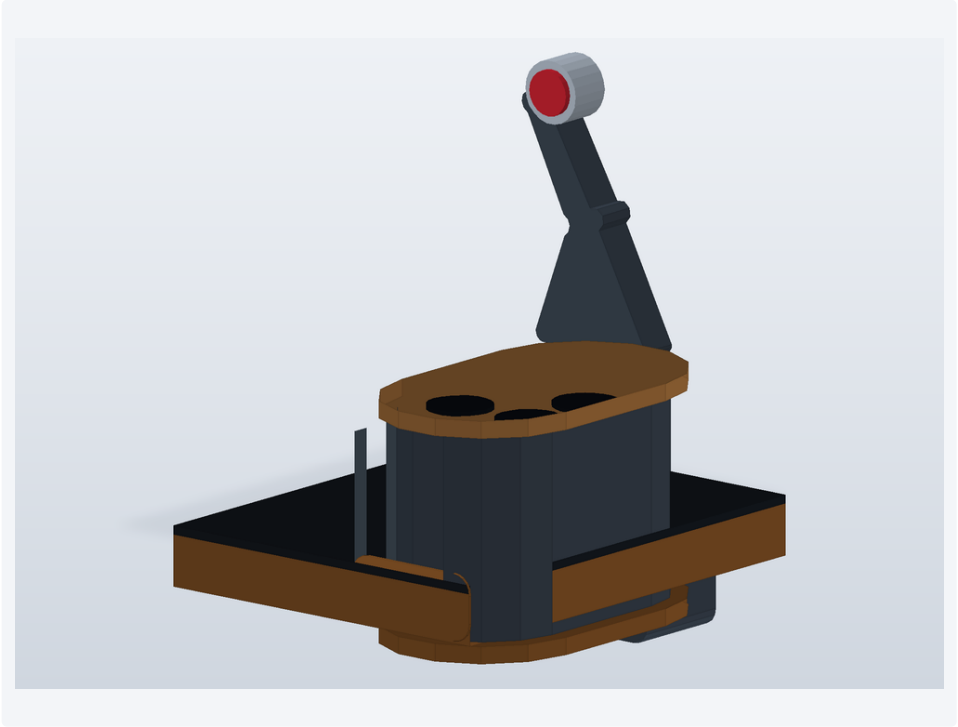
- 6 mm walnut
- 6 mm birch ply
- PETG

All dimensions millimetres. Part names match the exported STL / STEP files exactly.

PENS, DRAWER AND A MOUSE BUNGEE THAT CLEARS THE SWEEP

P4 · HexWing + Chimera Spire

A stadium-shaped walnut-and-black caddy clamps to the right edge at Z = 16...52 mm only, so the rest of the edge stays completely clear for your forearm. Six pen wells, a magnetic micro-drawer and a 94 mm airfoil spire that keeps the mouse cable off the mat.



HexWing at 40° / 14°: the clamp disappears behind the desk edge, exactly as designed.

7 PIECES IN THIS PRODUCT

W4A	Caddy floor	124 x 72 x 6	P4B	Fluted core	64 x 72 x 116
W4B	Pen collar	124 x 72 x 6	P4C	Drawer box	47.5 x 21.5 x 66
W4C	Drawer face	51.5 x 23.5 x 6	P4D	Airfoil spire	51.1 x 94.1 x 14
P4A	Right C-clamp	86 x 52 x 36			

WHAT IT DOES FOR YOU

- Six Ø23 mm wells hold pens, a stylus, an SD reader and a USB stick upright and findable.
- The micro-drawer swallows dongles, a spare tip and a microSD — the things you lose.
- The spire holds the mouse cable 94 mm up, so the cable never drags across the mat.
- Because the clamp is tucked to the rear 36 mm, nothing touches your forearm at the front edge.
- The whole caddy lifts off in one movement if you need the right edge back.

PUTTING IT TOGETHER

1. Clamp P4A over the right edge; it should stop with the walnut deck level and the jaw under the slab.
2. Screw W4A to the clamp with two M3 × 12 through the deck.
3. Drop P4B onto the deck, add W4B on top, then press P4D into its socket.

MATERIALS IN THIS PRODUCT

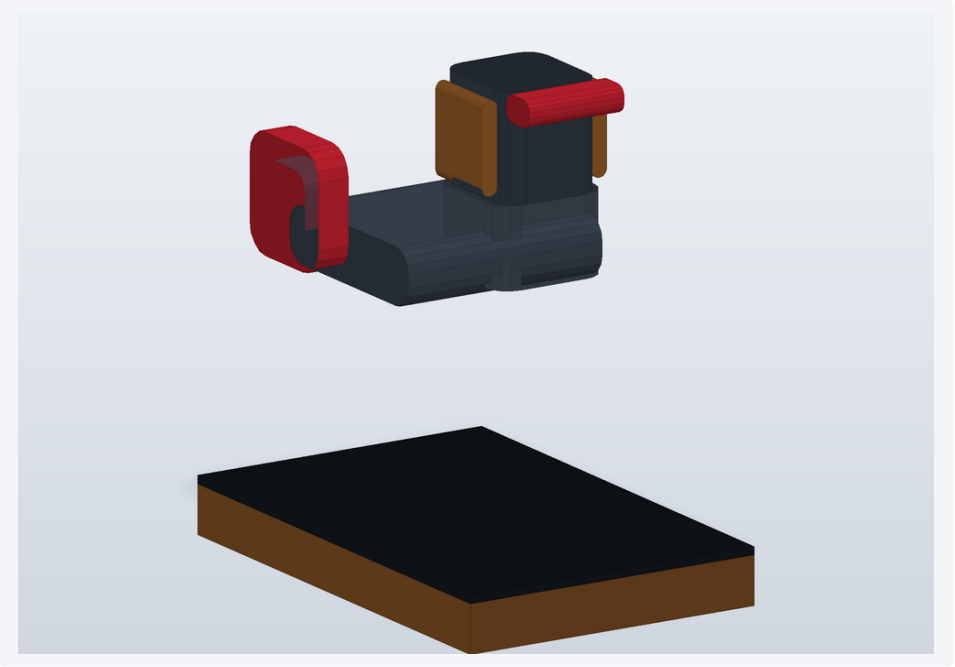
- 6 mm walnut
- PETG

All dimensions millimetres. Part names match the exported STL / STEP files exactly.

A HOLOGRAM IN A 25.4 MM CUBE

P5 · HoloBuddy

A 38 × 40 × 42 mm matte-black optical chamber holds a 1-inch 50:50 K9 beam-splitter cube, a 0.96 inch 128 × 64 OLED, an ESP32-C3 and a head-pat touch sensor. It mounts three ways: on the right DT-30 wing of the AeroBridge, on its own 608 pedestal, or freestanding on the desk.



HoloBuddy on its right-wing DT-30 mount: the wing arm slides into the red collar and the joint disappears.

9 PIECES IN THIS PRODUCT

W5A Cheek, left	34 x 30 x 6	P5B Electronics base	44 x 30 x 42
W5A Cheek, right	34 x 30 x 6	P5C Magnetic visor	38 x 10 x 14.7
W5B Pedestal base	Ø70 x 6	P5D DT-30 wing arm	70.7 x 51 x 76
G5 Beam-splitter	25.4 x 25.4 x 25.4	P5E Pedestal hub	36 x 9 x 36
P5A Optical core	38 x 40 x 42		

WHAT IT DOES FOR YOU

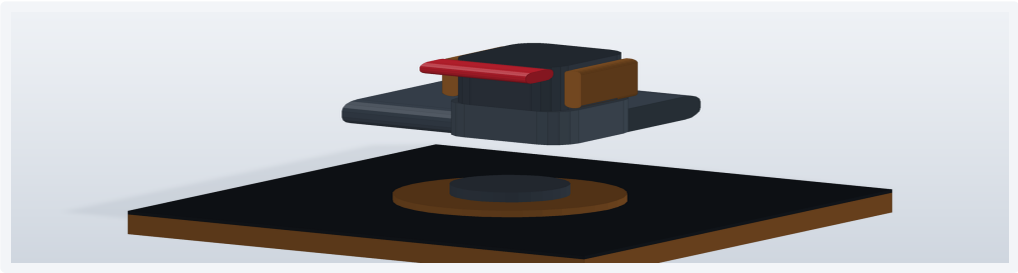
- Tap the top and the image flips or pauses — that is the TTP223 head-pat sensor.
- Speak and the INMP441 microphone picks it up; the ESP32 streams levels to the PC over USB or Wi-Fi.
- The floating image sits above the mouse sweep, so it never gets in the way of your hand.
- A magnetic brow visor snaps over the cube to kill glare in a bright room.
- Two walnut cheeks hide the seams and make it look like a piece of furniture.

PUTTING IT TOGETHER

1. Slide the DT-30 tongue on P5D into the right collar at X = 459 mm until it clicks.
2. Or press the 608 into P5E, drop W5B on the desk and set the head on the hub for the pedestal mount.
3. Load the ESP32 firmware, run the PC streamer, and the OLED image appears inside the cube.

MATERIALS IN THIS PRODUCT

- 6 mm walnut
- K9 optical glass, 50:50 coat
- matte-black PETG
- PETG



The pedestal mount: walnut coaster, printed 608 hub, head docks straight on it.

All dimensions millimetres. Part names match the exported STL / STEP files exactly.

Every wood piece — 14 parts

12 MM AND 6 MM SHEET

THE TWO STRUCTURAL ARCHES ARE 12 mm

PART	SIZE mm	MATERIAL	HOW TO MAKE IT	USED IN
W1A Left arch	184 x 131.4 x 12	12 mm walnut / birch ply	Band-saw or jigsaw, spindle-sand the C-curve, drill 2 x Ø8.1 rod bores	P1
W1B Right arch	184 x 131.4 x 12	12 mm walnut / birch ply	Mirror of W1A — flip the template, same tool path	P1
W1C Slat deck	360 x 182 x 6	6 mm walnut	Router 6 stadium slots, ease the front lip with an R3 round-over	P1

DECKS, TRAYS AND COASTERS — ALL 6 mm

PART	SIZE mm	MATERIAL	HOW TO MAKE IT	USED IN
W2A Vault floor	184 x 52 x 6	6 mm birch ply	Table-saw rectangle, 4 vent slots, glue-and-clamp to the shield	P2
W2B Vault shield	184 x 84 x 6	6 mm walnut	Nine 9 mm vent slots, 2 mm round-over on the outer face	P2
W2C ULT saddle	99 x 38 x 20	20 mm walnut block	Bandsaw crescent, cove the seat with a 38 mm core-box bit	P2
W3A Bottle coaster	Ø84 x 6	6 mm walnut	Ø84 hole saw, chamfer the rim, 2 coats oil	P3
W3B Notebook tray	172 x 162 x 6	6 mm birch ply	Jigsaw crescent scoop, 10 mm corner radius, sand to 320	P3
W4A Caddy floor	124 x 72 x 6	6 mm walnut	22 mm-radius stadium blank, 4 x M3 pilot holes	P4
W4B Pen collar	124 x 72 x 6	6 mm walnut	Same stadium blank, 6 x Ø23 wells, 2 mm round-over	P4
W4C Drawer face	51.5 x 23.5 x 6	6 mm walnut	Bandsaw, 7 mm radius, finger pull cove, glue to P4C	P4
W5A Cheek, left	34 x 30 x 6	6 mm walnut	6 mm radius corners, 2 mm round-over — one of a mirrored pair	P5
W5A Cheek, right	34 x 30 x 6	6 mm walnut	Mirror of the left cheek	P5
W5B Pedestal base	Ø70 x 6	6 mm walnut	Ø70 hole saw, Ø36 relief for the 608 hub, chamfer the rim	P5

Cut from 12 mm and 6 mm walnut / birch. All flat, all saw-and-drill work. Fixture and grain notes are on the cut-guide page.

Metal, glass and the first printed nodes

METAL AND OPTICS — 3 PARTS

PART	SIZE mm	MATERIAL	HOW TO MAKE IT	USED IN
M1A Rear tie-rod	Ø8.0 x 360	6061 aluminium / 304 stainless	Hacksaw or pipe-cutter, de-burr, 400-grit lengthwise for a brushed finish	P1
M1B Front tie-rod	Ø8.0 x 360	6061 aluminium / 304 stainless	Same cut, same finish	P1
G5 Beam-splitter	25.4 x 25.4 x 25.4	K9 optical glass, 50:50 coat	Buy — 1-inch 50:50 non-polarising cube, 400-700 nm	P5

3D-PRINTED NODES 1 OF 2 — 13 PARTS

PART	SIZE mm	MATERIAL	HOW TO MAKE IT	USED IN
P0 Fit coupon	90 x 150 x 14	PETG	Print first. Confirms your tolerances before anything else	P0
P1D Stopper L	44 x 27.5 x 27.2	TPU 95A	0.2 mm layers, 3 walls, 20% infill — one of a pair	P1
P1D Stopper R	44 x 27.5 x 27.2	TPU 95A	Mirror of the left stopper	P1
P1E DT-30 collar L	11 x 44 x 44	PETG	0.16 mm layers, 4 walls; dovetail root 30 mm at 60°	P1
P1E DT-30 collar R	11 x 44 x 44	PETG	Receives the HoloBuddy wing at X = 459 ± 0.15 mm	P1
P1F MagPivot arm	68.8 x 38 x 60	PETG	Print flat on its side, no supports; 608 bore reamed to 8.05	P1
P1G MagSafe disc	Ø70 x 9	PETG	8 x 6 x 3 N52 pockets press-fit at 54 mm PCD	P1
P2A Rear clamp arch	58 x 133.5 x 28	PETG	Print upright, 4 walls; 2 mm EVA pad recessed in each jaw	P2
P2B Front clamp arch	58 x 133.5 x 28	PETG	Identical part — the pair sets the 184 mm vault width	P2
P2C Cable spine	22 x 70 x 26	PETG	Screw-free snap into the shield, 22 mm cable channel	P2
P2D 608 bearing hub	50 x 47.5 x 48	PETG	608 bore 22.15 mm interference; print with 5 perimeters	P2
P2E Axle collar	24 x 13.4 x 24	PETG	Cyber-Red accent, M4 grub screw clamps the swing arm	P2
P3A Bottle chalice	99 x 186.5 x 121.5	PETG	Largest node — 186.5 mm tall, fits a 200 mm bed. Print upright with a brim	P3

The last 13 printed nodes

3D-PRINTED NODES 2 OF 2 — 13 PARTS

PART	SIZE mm	MATERIAL	HOW TO MAKE IT	USED IN
P3B Left runner	24 x 58 x 48	PETG	Clamp body + DT-30 slide rail in one part	P3
P3B Right runner	24 x 58 x 48	PETG	Runs the 110 mm tray travel	P3
P3C Slide cleat L	14 x 6 x 146	PETG	Glue or screw to the walnut tray underside	P3
P3C Slide cleat R	14 x 6 x 146	PETG	Mirror of the left cleat	P3
P4A Right C-clamp	86 x 52 x 36	PETG	Tucked to Z = 16...52 mm so the deck edge stays clear	P4
P4B Fluted core	64 x 72 x 116	PETG	6 x Ø23 pen wells, micro-drawer bay, 7 flutes	P4
P4C Drawer box	47.5 x 21.5 x 66	PETG	Print walls upright, 0.16 mm layers for the slide fit	P4
P4D Airfoil spire	51.1 x 94.1 x 14	PETG	Print flat, 14 mm thick; 608 light-pipe at the tip	P4
P5A Optical core	38 x 40 x 42	matte-black PETG	Kill the gloss: 0.12 mm layers, matte filament, no supports	P5
P5B Electronics base	44 x 30 x 42	PETG	ESP32-C3 cradle, INMP441 port, 608 swivel boss	P5
P5C Magnetic visor	38 x 10 x 14.7	PETG	Cyber-Red brow, 2 x N52 magnets, snaps on or flicks off	P5
P5D DT-30 wing arm	70.7 x 51 x 76	PETG	30 mm dovetail tongue slides into P1E at X = 459 mm	P5
P5E Pedestal hub	36 x 9 x 36	PETG	608 press seat for the freestanding desktop mount	P5

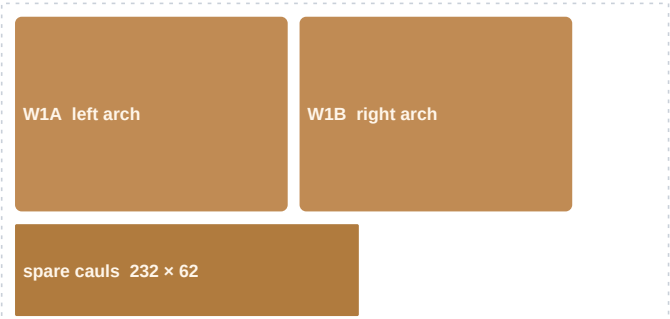
PRINT SETTINGS THAT MATTER

0.2 mm layers general · 0.16 mm on collars, hubs and the DT-30 joints · 4 walls · 25 % infill on structural nodes, 15 % on cosmetic parts · no supports anywhere except P4D.

Largest single node is P3A at 186.5 mm, so every part fits a 200 x 200 x 200 mm bed.

Three stock items cover every wood and metal part

SHEET A — 12 mm WALNUT / BIRCH, 450 × 220 mm



Both arches are 184 × 131.4 mm. They nest side by side, leaving 65 mm of spare stock at the right edge and 80 mm along the front for the cauls.

BAR B — Ø8.0 mm ROD, 1000 mm STOCK



Two straight cuts. De-burr, then draw the rod through 400-grit lengthwise for a brushed finish.

SHEET C — 6 mm WALNUT / BIRCH, 600 × 400 mm



All eleven 6 mm parts fit one 600 × 400 sheet with roughly 20 % spare. The two discs are the coaster and the pedestal base; cut them with a hole saw.

WHAT TO BUY

WALNUT / BIRCH PLY
12 mm sheet and one 600 × 400 × 6 mm offcut

ROD
2 × 360 mm of Ø8 mm 6061 or 304

FILAMENT
330 g PETG plus about 40 g TPU 95A

OPTICS & ELECTRONICS
K9 cube, 0.96 inch SSD1306, ESP32-C3, TTP223, INMP441

HARDWARE
2 × 608-2RS, 6 × 3 mm magnets, M3 / M4 inserts, 2 mm EVA

Sheet outlines are nested, not to exact scale. Allow a 3 mm kerf and 6 mm between parts.

The order matters: print the coupon, then the joints, then everything else

1. **Print P0 and test-fit the hardware**
   60 min print. Press in a 608, drop in a magnet, slide a DT-30 tongue. Adjust your slicer, not the model.
2. **Cut and finish the wood**
   Rip the 12 mm arches, then route the 6 mm sheet. Sand to 320 and oil everything before assembly — it is far easier flat.
3. **Cut the two rods**
   Two 360 mm lengths. De-burr both ends; a burr will not pass through the Ø8.1 bores in the arches.
4. **Print the structural nodes**
   P1E collars, P2A/P2B clamp arches, P2D hub, P3B runners, P4A clamp first — these set every fit. 4 walls, 0.16 mm layers.
5. **Assemble each product on the bench**
   Build P1 through P5 one at a time and check each on the desk before starting the next. Every product is independent.
6. **Fit the magnets and bearings**
   Magnets last: check polarity with a marker dot before you press them in. A reversed magnet cannot be removed.
7. **Set the clamp pads**
   2 mm EVA on every jaw face. This is what keeps the desk unmarked, and the only thing you should ever replace.
8. **Load the HoloBuddy firmware**
   Flash the ESP32-C3, run pc_holo_streamer.py, then calibrate the 45° mirror so the image sits upright.

TOOLKIT — NOTHING EXOTIC

Jigsaw or band saw · Ø84 and Ø70 hole saws · 6 mm router bit · 3 / 4 / 8 mm drill bits · Torx and hex keys · 320-grit paper · Danish oil · a 3D printer with a 200 mm bed. No CNC, no laser and no sheet-metal brake is required anywhere in this build.

A normal working day with the system

08:40 — **Sit down**
The keyboard comes out of the garage, the mouse is already on a clear mat. HoloBuddy shows the time and the first meeting.

09:15 — **Laptop docked**
The G16 sits at 18° with its intakes open. The phone tilts up on the MagPivot for a video call.

11:00 — **Water**
The bottle goes into the chalice, outboard of the desk. It cannot spill onto the laptop because it is not on the desk.

13:30 — **Lunch**
Headphones swing out on the 608, then swing back under. The notebook tray pulls out for notes.

15:00 — **Heads-down**
Pens and the dongle come out of the HexWing drawer. The spire holds the mouse cable off the mat.

17:30 — **Pack down**
Keyboard into the garage, lid closed, phone off the arm. The desk is clear in under a minute.

MAINTENANCE Re-oil the wood once a year · brush the slat slots monthly · replace the EVA pads when they compress · re-flash the ESP32 only if you change the optics.

Files, names and how this book stays true

WHAT IS IN THE REPOSITORY

Product0_FitCheckGauge/
Fit coupon — build_product() plus its .manifest

Product1_AeroBridgeG16/
Bridge, deck, collars, MagPivot arm and disc

Product2_StealthVaultULT/
Vault arches, shield, cable spine, bearing hub and saddle

Product3_HydroSling/
Chalice, coaster, runners, cleats and notebook tray

Product4_HexWingSpire/
Right clamp, fluted core, drawer and airfoil spire

Product5_HoloBuddy/
Optical core, electronics base, visor, wing and pedestal

Master_FullDeskSetup/
The whole desk in one script: exports STL, 3MF, STEP + CSV manifest

HoloBuddy_Firmware_And_PC/
ESP32 firmware, PC streamer, wiring and optics guide

verify_fusion_scripts.py
Rebuilds every model headless and checks 15 geometric rules

THE CHECKS THIS BOOK DEPENDS ON

- ✓ Every part fits a 200 mm bed (max 186.5 mm)
- ✓ Nothing penetrates the back wall (Z ≥ 0)
- ✓ The vault and the headphone knuckle never overlap
- ✓ The HoloBuddy tongue seats at X = 459 mm
- ✓ The keyboard garage clears both arches
- ✓ The 165 × 230 mm mouse sweep is clear
- ✓ All 41 parts form 6 desk-anchored groups
- ✓ The 415 mm knee void is clear
- ✓ No part floats free of the assembly

ON THE REFERENCE HARDWARE

The hardware in this build (ROG Strix G16, AULA F75, Kreo Chimera V2, Sony ULT WEAR, Samsung S24+, ROG 280 W brick) was specified in the brief. The amzn.to shortlinks in that brief could not be opened from the build environment, so no link-dependent specification is baked into any model. Every camera, port and pocket is driven by a named Fusion user parameter, so if a device changes you edit one number and rebuild.